Virtual Skirts: Floor-Edge Vortices for Underbody Sealing on a Formula Student Car

Jordi Facha Álvarez
Aerodynamics Department, UFV Racing Team
Universidad Francisco de Vitoria, Madrid, Spain

Abstract — Formula Student cars rely on underbody downforce at low speeds and small ride heights, but air drawn in at the floor edges reduces the suction beneath the floor, and the rules prevent the gap from being sealed with physical skirts. This paper examines how streamwise vortices along the floor edges, known as virtual skirts, could seal the floor of the UFV Racing Team car, which runs at a typical speed of 48 km/h and at the minimum ride height of 30 mm. After reviewing ground-effect aerodynamics, the history of floor sealing in motorsport and the formation of edge vortices, it argues that the rounded channel on the current floor edge is unlikely to create a strong vortex on its own. Two proposals are developed that leave the ride height unchanged: vortex-generating vanes on the upper surface of the existing floor, which set the direction of spin and make the vortex drift toward the edge, and a new floor edge with a gradual curved rise ending in a sharp lip, which fixes the vortex at the gap. The two are shown to be complementary, and a staged plan is recommended: evaluate both in CFD, fit the low-cost vanes to the current car if they prove effective, and build the sharp lip into the next floor design.

Keywords — ground effect, underbody, edge vortex, virtual skirt, Formula Student.

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1. Introduction

Formula Student (FS) is an international engineering competition in which university teams design, build and race a small single-seat, formula-style car. The racing takes place on short, twisty courses: the rules (D 6.2, D 7.1) limit straights to 80 m and require slaloms, chicanes and hairpins, with turns as tight as 9 m in diameter[1]. As a result, an FS car spends most of each lap cornering at low to moderate speed.

On a course like this, lap time depends mostly on how quickly the car can take corners, and that is limited by the grip of its tires. One way to increase grip is *downforce*: an aerodynamic force that pushes the car down onto the track, pressing the tires harder into the surface without adding any mass[2],[3]. This makes downforce valuable even at the modest speeds of FS. Throughout this paper, a reference speed of 48 km/h (13.3 m/s) is used. This is the team's standard speed for aerodynamic analysis and is representative of the speed at which the car spends most of its time on track.

Of all the parts of the car that produce downforce, the *floor* (also called the *underbody*) is of particular interest. A wing produces downforce by deflecting air upwards. The floor works differently: it uses the ground. Air that enters the narrow gap between the floor and the track is forced to speed up, and faster-moving air has a lower pressure. The pressure under the car therefore drops below the pressure acting on its upper surfaces, and this difference pulls the car toward the ground[2],[4].

Because the floor is an almost flat surface aligned with the oncoming air, it generates this force with comparatively little *drag*, the air resistance that slows the car down. This makes it one of the most efficient sources of downforce on a race car[2],[4]. Its performance, however, is very sensitive to two things: the *ride height*, meaning the distance between the floor and the ground, and what happens to the air along the floor's edges. This paper is about the second.

The problem at the edges is leakage. The air beside the car is at a higher pressure than the

air beneath it, so it is drawn sideways into the gap under the floor edge. This raises the pressure under the car and reduces the downforce[4]. In the late 1970s, Formula One teams solved this mechanically with *sliding skirts*: panels along the sides of the car that rubbed along the track and closed the gap completely[2]. The FS rules forbid any aerodynamic device that touches the track (T 2.2.2) and require at least 30 mm of ground clearance (T 2.2.1)[1], so an open gap at the floor edge cannot be avoided.

The alternative is to seal the gap with air instead of a solid wall. A *vortex*, a tube of rapidly rotating air, running lengthwise along the floor edge can act as a barrier that blocks the air trying to flow in. Because it seals the floor without touching the track, this effect is commonly described as a "virtual skirt"[5].

The UFV Racing Team car has a flat floor that ends in a *diffuser*, an upswept section at the rear whose role is explained in Section 2. Along the outer edge of each side of the floor, from just behind the front wheel to just ahead of the rear wheel, runs a smooth concave channel about 25 mm in diameter, open toward the ground. Its exact purpose is not documented: a curved edge of this kind stiffens the thin floor panel and protects it from damage, and it may also be intended to hold an edge vortex in place. On its own, however, a smooth, rounded channel is unlikely to create a strong vortex, because vortices form most readily where air is forced to separate from a sharp edge (Section 5). Something must create the vortex first.

This paper therefore investigates how vortex generators, small devices designed specifically to create vortices, could be added to the current floor so that a vortex forms along each edge and seals the underbody, with the existing channel helping to hold it in place. It also briefly explores an alternative in which the channel is removed and the edge is sealed by vortex generators alone, while the stiffness the channel provides is kept by other means. In both cases, the design must account for the rest of the car. The turbulent wake shed by each front tire flows over the start of the floor edge and can weaken a vortex before it is established, while a vortex that strikes a rear tire would break down and disturb the flow around the tire and the diffuser. The path of the vortex must therefore be controlled along the full length of the floor.

The study draws on published wind-tunnel and simulation research, established textbooks and the competition rules; it presents no new simulation or test data. Section 2 explains how a flat floor with a diffuser produces downforce, and Section 3 describes how leakage at the edges reduces it. Section 4 reviews how motorsport has dealt with this problem. Section 5 introduces how vortices form, and Section 6 describes the main types of vortex generator and how an edge

vortex can seal a floor. Section 7 discusses the constraints specific to Formula Student, including the influence of the front and rear tires. Sections 8 and 9 apply these findings to the team's floor: Section 8 proposes vortex generators for the current design, and Section 9 briefly explores the alternative without the channel. Section 10 presents the recommendation and a plan for testing it, and Section 11 concludes.

2. How a Flat Floor Generates Downforce

To understand how a floor pulls a car toward the ground, it helps to start with one of the basic principles of fluid flow: in a smooth flow, speed and pressure are linked. Where air speeds up, its pressure falls; where it slows down, its pressure rises. This relationship is described by Bernoulli's equation[3]:

$$p + \tfrac{1}{2}\rho V^2 = \text{constant} \qquad (1)$$

Here p is the *static pressure*, the pressure the air exerts on any surface it flows along; ρ (rho) is the density of the air, about 1.225 kg/m³ at sea level; and V is the local speed of the flow. The term $\tfrac{1}{2}\rho V^2$ is called the *dynamic pressure* and is usually written q. It can be thought of as the extra pressure the air would exert if it were brought completely to rest against a surface. Strictly, Eq. (1) holds only where friction in the air is negligible and the air does not noticeably compress. Both conditions are met by the main flow beneath an FS car, away from the thin layers of slow air that cling to every surface[2],[3].

Because the pressures around a car change with its speed, aerodynamicists usually describe them with a number that does not depend on speed, the *pressure coefficient*[3]:

$$C_p = (p - p_\infty) / (\tfrac{1}{2}\rho V_\infty^2) \qquad (2)$$

where p_∞ and V_∞ are the pressure and speed of the undisturbed air far ahead of the car. A value of $C_p = 0$ means the local pressure equals that of the surrounding atmosphere. $C_p = 1$ is the highest possible value and occurs at a *stagnation point*, where the air is brought to rest, such as the very tip of the nose. Negative values mean suction. Combining Eqs. (1) and (2) gives a simple and useful result: $C_p = 1 - (V/V_\infty)^2$. Wherever the air moves faster than the car itself, C_p is negative. Air flowing beneath the floor at 1.5 times the car's speed, for example, has $C_p = -1.25$.

Beneath a car, the floor and the track form a channel through which air must pass. When

the floor is close to the ground, the air entering this channel speeds up and, by Eq. (1), its pressure drops below that of the surrounding air[3],[4]. The upper surfaces of the car, meanwhile, see pressures much closer to atmospheric. The downforce produced by the floor is the sum of these pressure differences over its whole area, so even a small pressure difference produces a large force when it acts over a large surface. This is why the floor, which covers most of the car's plan-view area, is such a powerful source of downforce. The increase in aerodynamic force caused by the nearby ground is known as *ground effect*[4].

The strength of ground effect depends strongly on ride height. Wind-tunnel experiments on simplified car bodies fitted with a diffuser show that, as the floor is lowered toward the ground, downforce first rises steadily, then reaches a maximum, and finally drops sharply if the floor is lowered further[4],[6]. Zhang *et al.* describe these as three regimes: *downforce enhancement*, *maximum downforce* and *downforce reduction* (Figure 1)[4]. Figure 1 illustrates three cases. At A, the floor is too low: the flow beneath it is choked and the vortices break down. At B, close to the optimum, the suction is strongest. At C, the floor is too high: air passes underneath easily and little suction is produced. The drop at very low ride heights is linked to two effects examined in Sections 5 and 6: the air separating from the diffuser surface, and the breakdown of the vortices that form along the diffuser's edges[6]. In other words, edge vortices are not only a possible way of sealing the floor; they are already part of how a floor with a diffuser produces downforce.

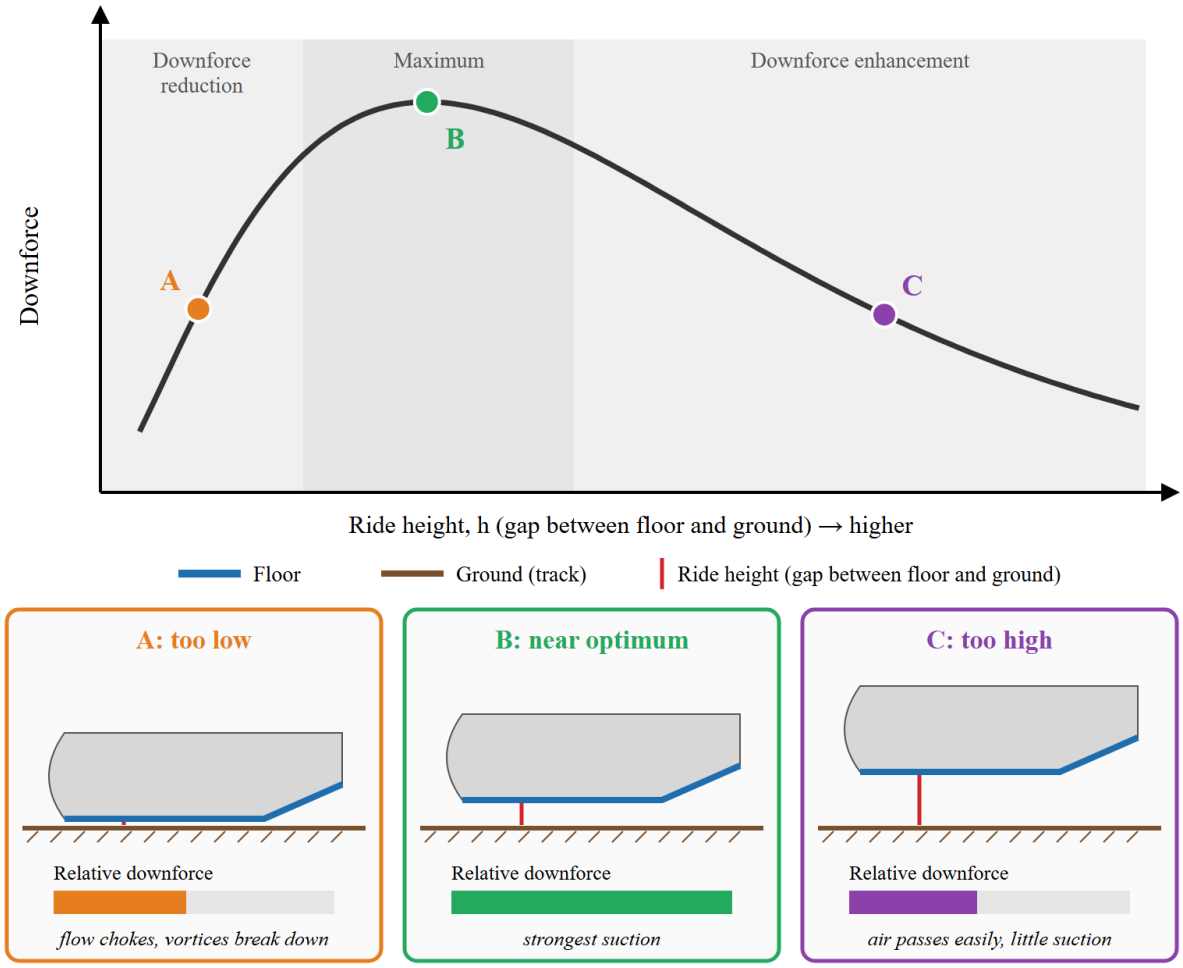


Figure 1. Typical variation of downforce with ride height, showing the three regimes described by Zhang *et al.* [4]. Illustrative only; not to scale.

For an FS car, the lowest allowed ride height is set by the 30 mm ground clearance rule[1]. Where this falls on the curve of Figure 1 depends on the length and shape of the floor and diffuser, so the curve cannot be read directly for any particular car. It does show, however, why small changes in ride height, caused by suspension movement under braking, cornering or bumps, can noticeably change the downforce a floor produces.

The diffuser is what makes this whole process work. It is an upswept section at the rear of the floor in which the gap between the floor and the ground gradually widens (Figure 2a). As the passage widens, the air slows down and its pressure rises again, a process called *pressure recovery*[2],[4]. Why does this lower the pressure? Two facts combine. First, the same amount of air must pass through every cross-section of the passage each second, so where the passage is

narrow the air must move quickly, and where it is wide it can move slowly. Second, at the diffuser exit the air rejoins the flow behind the car, so its pressure there is fixed at roughly atmospheric. Traveling from the narrow entrance of the diffuser, known as the *throat*, to the wide exit, the air slows down and, by Eq. (1), its pressure rises. If the pressure rises all the way through the diffuser and must finish at atmospheric, it has to start below atmospheric. A useful picture is a staircase whose top step is fixed at ground level: the more steps the diffuser climbs, the further below ground the bottom step must be. The lower pressure at the throat in turn draws air through the gap beneath the car more quickly, reducing the pressure along the whole floor[4].

A diffuser can only slow the air down gradually. If it is too steep, the slowing air can no longer follow the upswept surface and detaches from it, an effect called *flow separation*, and much of the pressure recovery is lost. Ruhrmann and Zhang tested a simplified body with five different diffuser angles and found that the steeper diffusers suffered from flow separation and vortex breakdown, which limited the downforce they could produce[6].

The result is a pressure that varies along the length of the floor, as shown schematically in Figure 2b. Suction builds up along the flat floor and peaks at the throat, where the air moves fastest; it then falls through the diffuser as the pressure recovers[4]. A smaller suction peak typically appears near the front, where the air accelerates around the leading edge of the floor. This distribution matters for the rest of this paper: the region where the pressure under the floor is lowest, around the throat, is also where outside air is most strongly drawn in from the sides. Section 3 examines this in detail.

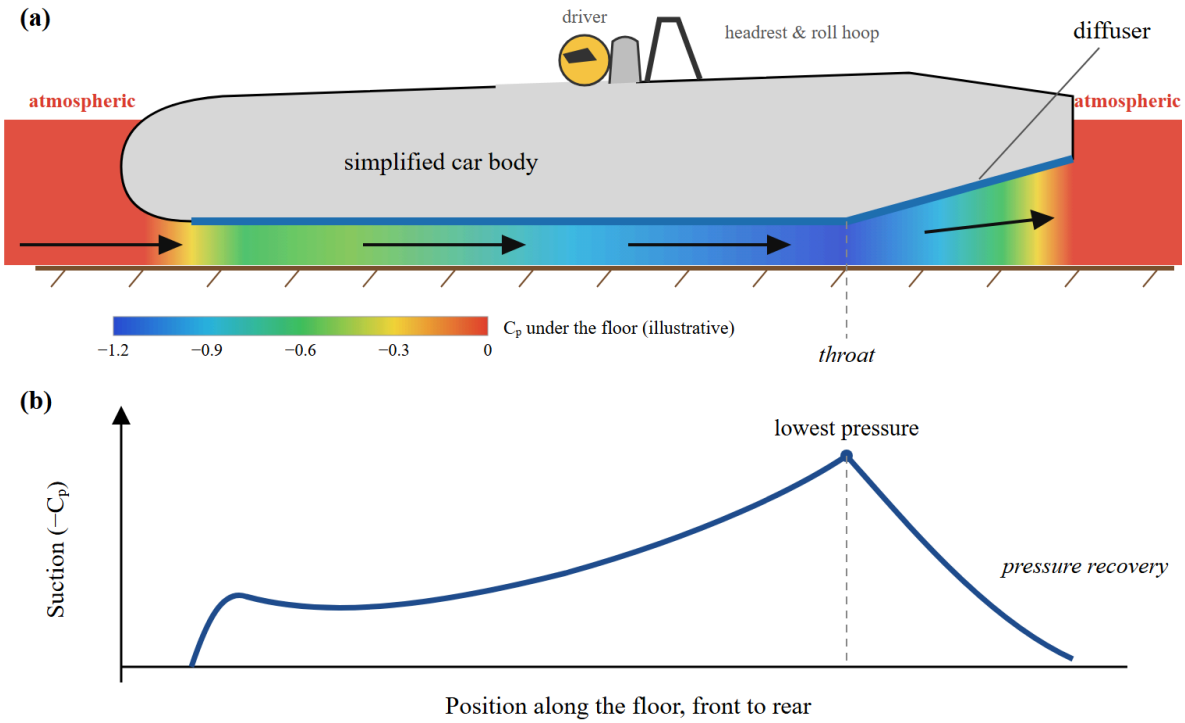


Figure 2. (a) Side view of a car with a flat floor and rear diffuser, with the pressure beneath it. (b) Suction along the floor, peaking at the throat. Illustrative only.

Finally, it is useful to put numbers to these ideas at the reference speed of 48 km/h (13.3 m/s). With ρ = 1.225 kg/m³, the dynamic pressure is $q = \frac{1}{2}\rho V^2 \approx 109$ Pa, roughly one thousandth of atmospheric pressure (101,325 Pa). Every pressure difference around the car is a fraction or a small multiple of this value, which shows how small the pressures that generate downforce really are. To give the numbers in this paper a consistent scale, a simple reference floor is used throughout: a flat floor with a plan area of 1.5 m² and an average pressure coefficient of −1, at 48 km/h. It is an illustrative value for quick, order-of-magnitude comparisons only; it does not represent the UFV floor and ignores the detailed effects discussed in later sections. The reference floor produces about 163 N of downforce, roughly the weight of a 17 kg mass. Because q grows with the square of speed, the same floor would produce four times as much at 96 km/h.

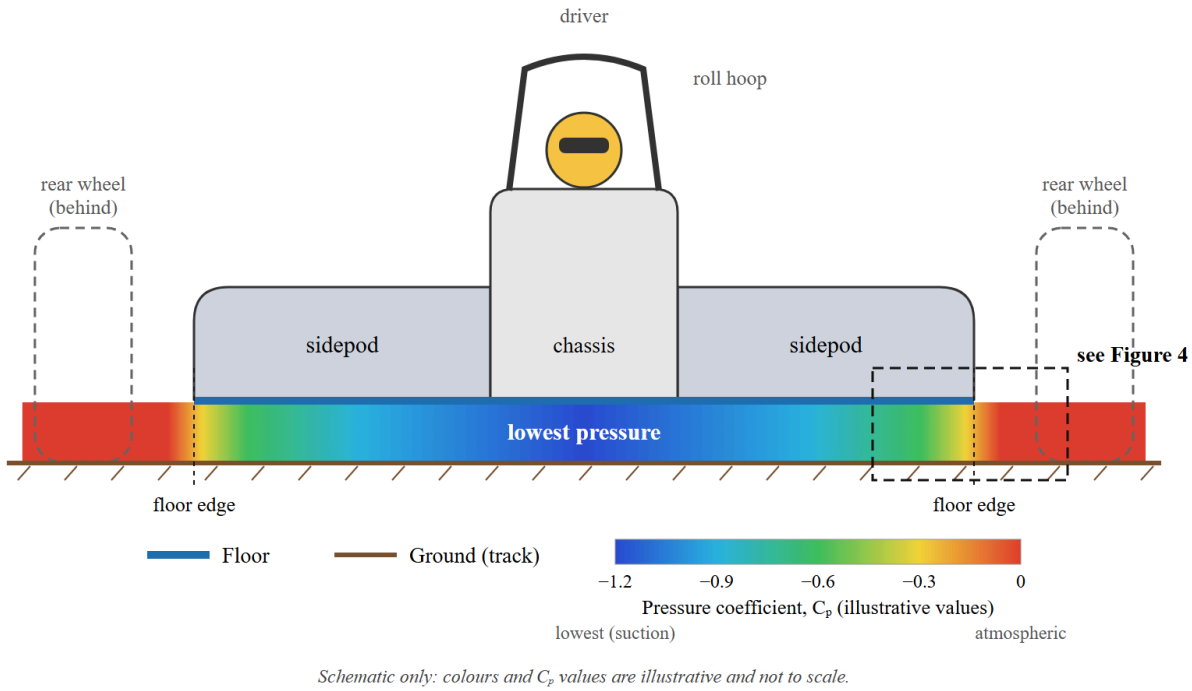
The flow is also described by the *Reynolds number*, $\mathrm{Re} = \rho VL/\mu$, where L is a characteristic length and μ is the viscosity of air, about 1.8×10^{-5} Pa·s. The Reynolds number compares the momentum of the moving air with its internal friction. At high values, momentum dominates and the slow layers of air next to the surfaces, called *boundary layers*, stay thin relative to the car; at lower values, friction plays a larger role, boundary layers are relatively thicker and the flow separates more easily[3]. Taking L as the measured length of the UFV floor,

0.94 m (Section 8), gives Re ≈ 8.5×10^5 at 48 km/h, more than an order of magnitude lower than for a Formula One car at racing speed. What this means for edge vortices is discussed in Section 7.

3. The Problem: Air Leaking in at the Sides

Section 2 showed that the floor works by keeping the pressure beneath it below the pressure of the surrounding air. At the sides of the car, this arrangement is difficult to maintain. Just outside the floor edge, the air is close to atmospheric pressure; a few centimeters inboard, beneath the floor, it is well below (Figure 3). Air always flows from high pressure toward low pressure, so where the two regions meet, at the gap between the floor edge and the track, air is pushed sideways into the space beneath the car[3],[4]. A helpful picture is a packed crowd next to a half-empty area: open a gap in the barrier between them and people stream through. Nobody in the empty area pulls them in; they are pushed by the crowd behind them. Air behaves in the same way. Its molecules constantly collide with and push on one another, so where the pressure is lower, air from the higher-pressure side is pushed in. Strictly, then, the floor does not suck air in; the surrounding air pushes it in. This is the leakage introduced in Section 1. Because the FS rules make it impossible to close this gap (T 2.2.1, T 2.2.2)[1], some leakage is present on every car.

Cross-section through the car, seen from the front, with the pressure under the floor



Schematic only: colours and C_p values are illustrative and not to scale.

Figure 3. Front cross-section of the car showing the pressure beneath the floor: lowest at the center, rising to atmospheric beyond the floor edges. Illustrative only.

The leaking air reduces downforce in a simple way: it adds to the air already beneath the floor and raises its pressure, partly filling in the suction that the floor depends on[2],[4]. Because the air enters at the sides, the effect can be expected to be strongest close to the edges and to weaken toward the center line, so that the outer part of the floor produces less downforce than the middle. In effect, the floor then behaves as if it were narrower than it really is. For the reference floor of Section 2, if leakage halved the suction over the outer fifth of its area, its downforce would fall by about a tenth, from roughly 163 N to 147 N.

Leakage is also uneven along the length of the car. It is driven by the pressure difference between the air outside and the air beneath the floor, and Section 2 showed that this difference is largest around the diffuser throat (Figure 2b). The strongest inflow therefore occurs in the rear part of the floor, just ahead of the diffuser and along its sides. This is precisely where the floor produces most of its downforce, so it is where leakage costs the most. It is also the region closest to the rear tires, a point that returns in Section 7.

The leaking air does not simply flow straight in. As it passes around the floor edge, it is turned sharply and begins to rotate, rolling up into a vortex that runs lengthwise along the

underside of the floor (Figure 4). Wind-tunnel experiments on a simplified body with a diffuser have observed exactly this: air separating from the lower side edges of the body rolled up into vortices along each side of the diffuser[7]. These naturally formed vortices were not harmful in themselves. Mahon *et al.* found that they increased downforce as the ride height was reduced until, at very low ride heights, they broke down, which reduced downforce and increased drag[7].

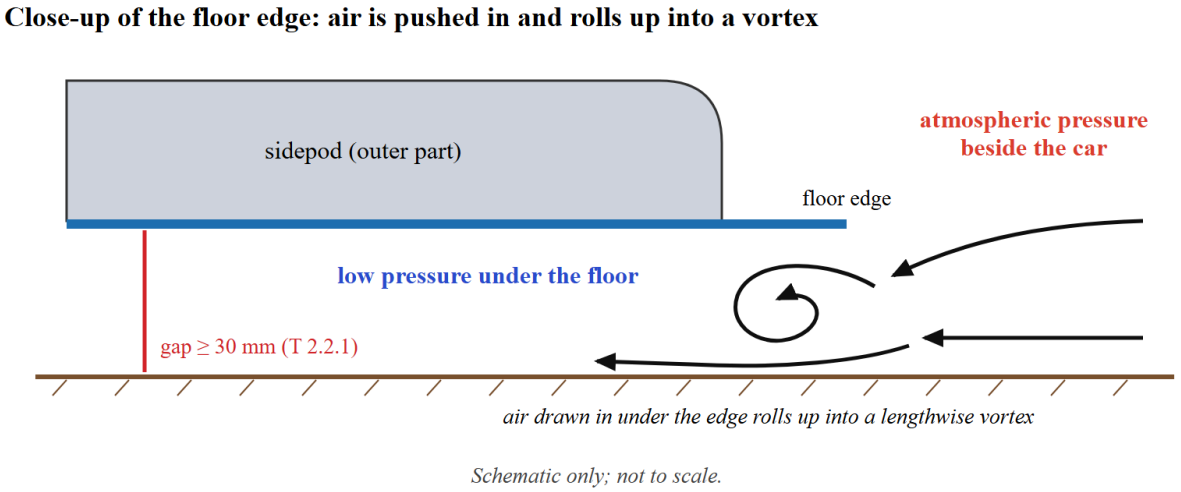


Figure 4. Close-up of the floor edge, where inflowing air rolls up into a vortex. Illustrative only.

This observation is central to the rest of this paper. Some rotation will form at the floor edge whatever the designer does; the real question is whether it is weak and left to chance, or strong and placed where it is useful. A vortex that is strong enough and in the right position can act as the barrier described in Section 1, blocking much of the inflow. A weak or badly placed one does little, or breaks down. Creating and controlling this vortex deliberately is the idea behind the virtual skirt, and the purpose of the vortex generators examined in Sections 6, 8 and 9.

Finally, leakage changes as the car moves. In a corner, the car travels at a small angle to the oncoming air, known as yaw, so the air no longer arrives straight on. One side of the floor then faces the oncoming flow more directly than the other, so the pressure around the two edges, and with it the leakage, can be expected to become unequal. An edge seal must therefore keep working when the car is yawed, not only when it travels in a straight line. This, together with the influence of the tires, is discussed in Section 7.

4. How Motorsport Has Sealed the Floor

The value of sealing the floor edges was first demonstrated in Formula One in the late

1970s. The Lotus 78 of 1977 shaped the underside of its sidepods to generate suction and used skirts along its sides to seal in the low pressure. Its successor, the Lotus 79, refined the idea with improved underbody tunnels and a diffuser, and dominated the 1978 season[8]. The skirts rubbed along the track, closing the gap at the floor edge mechanically so that leakage was almost eliminated. Rival teams quickly copied the concept.

The skirts also revealed the risk of depending on a perfect seal. The downforce of these cars was never reliably constant: the low-pressure region could shift and set the car bouncing, an effect known as porpoising, and if a skirt came loose or broke, much of the downforce could disappear at once[8]. Combined with greatly increased cornering speeds, this made the cars hard to control. For 1981, the governing body banned skirts and imposed a 6 cm minimum ground clearance. Skirts were allowed back in 1982, but after a season marked by several serious accidents, flat floors between the tires became mandatory from 1983 and skirts were banned for good[8]. The FS rules reflect the same concern: aerodynamic devices that touch the track are prohibited (T 2.2.2)[1].

A more radical idea appeared in 1978 on the Brabham BT46B, which used a large fan at the rear of the car to pull air out from under the chassis and create very high downforce. The car won the only championship race it entered, after which the team withdrew it[9]. This route is effectively closed to FS teams, because the rules limit the combined power of all devices that move air, including cooling fans, to 500 W (T 11.11.1)[1].

Once flat floors were mandatory, teams turned to the airflow itself to protect the underbody. A particular threat came from the rear tires. Where a rotating tire meets the ground, it displaces air sideways in fast jets known as tire squirt[10],[11], and the jet on the inboard side flows toward the edge of the diffuser and disturbs the flow inside it. Red Bull's 2010 RB6, designed by Adrian Newey, used engine exhaust to counter this. Its exhaust outlets were moved down so that they blew over the diffuser, and the jet rolled around the diffuser's side walls, keeping it sealed[10]. A later layout blew the exhaust into the channel between the side of the tire and the outer wall of the diffuser, specifically to reduce the effect of tire squirt[10]. The rules restricted exhaust positions from 2011 and effectively ended the practice in 2012[10]. Unlike skirts and fans, exhaust blowing is not ruled out for a combustion-engined FS car such as the UFV car: the rules only require the outlet to point to the side or rear of the car, to end no more than 450 mm behind the rear axle and to sit no higher than 600 mm above the ground (CV 3.1.1, CV 3.1.2). As discussed in Section 7, however, it is impractical for an FS car.

Modern Formula One floors rely instead on vortices. Anderson's analysis of the 2021 Aston Martin floor describes a row of curved vertical vanes along the floor edge. The first vanes set up vortices at the front corner of the floor; the following ones turn air from above the floor outboard into those vortices to strengthen them; and the rearmost guide the flow so that it joins the tire squirt outside the rear tire, all with the aim of reducing the air drawn under the car[11]. Since full ground-effect floors returned to Formula One in 2022, vortices running along the floor edges have been described as virtual skirts that maintain the low pressure beneath the car[5].

Three lessons follow for an FS car. First, sealing the floor edges is worth pursuing: Formula One teams have spent almost fifty years finding ways to do it. Second, the most direct solutions are banned or heavily limited under FS rules: skirts are prohibited and fans are restricted to 500 W, while exhaust blowing, although permitted for combustion cars, is impractical, as discussed in Section 7. Third, the approach that is fully open to every team is the one that leading teams use today: shaping the floor edge and adding small passive devices so that the air itself forms the seal. The Formula One examples also show that creating a vortex is not enough; it must also be positioned and steered, in particular around the rear tire. These are exactly the requirements identified in Section 1 for the UFV floor.

5. What a Vortex Is and How It Forms

A vortex is a region of flow in which the air rotates around a common axis: an imaginary line through its center, like the axle of a wheel. Everyday examples include the swirl of water draining from a bath and the funnel of a tornado; in aerodynamics, the most familiar example is the vortex trailing from each wingtip of an aircraft, sometimes made visible by condensation in humid air. The vortices of interest in this paper are streamwise vortices: their axis of rotation points in the direction of travel. Seen from the front of the car, the air spins in a circle, while the vortex as a whole streams rearwards along the floor edge, like a corkscrew lying on its side.

A vortex has two distinct regions (Figure 5). At its center is the core, a narrow region that rotates almost like a solid body: the rotational speed rises from zero on the axis to a maximum at the edge of the core. Outside the core, the rotational speed falls steadily with distance. Because the air in a vortex travels in circles, it must be pushed continuously toward the center, and that push comes from a pressure difference: the pressure is lowest on the axis and rises toward the outside[12]. The core of a strong vortex is therefore a line of low pressure. When such a vortex runs close to the underside of a floor, its low pressure adds to the suction already produced by the

floor, which helps to explain why the edge vortices observed by Mahon *et al.* increased downforce[7].

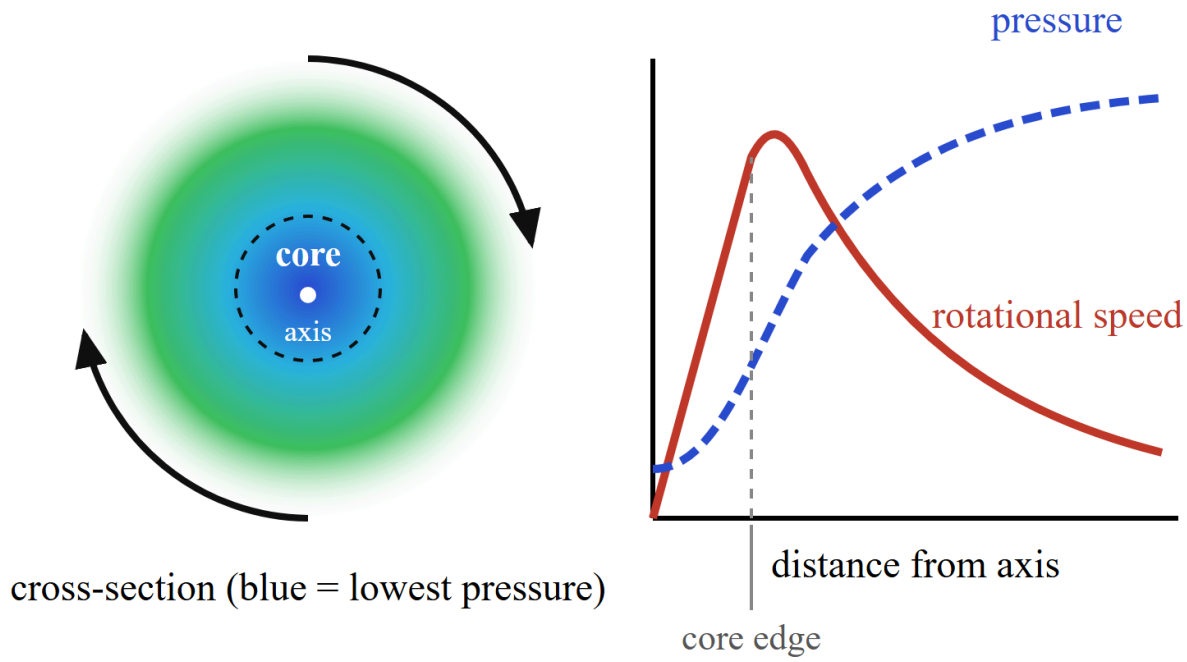


Figure 5. Structure of a vortex: rotational speed peaks at the edge of the core, and pressure is lowest on the axis. Schematic only.

Vortices form wherever air flows around an edge from a region of high pressure into a region of low pressure. At a wingtip, the higher-pressure air beneath the wing flows around the tip toward the lower pressure above it, and as the wing moves forward this sideways flow rolls up into a trailing vortex. A floor edge is the same situation turned upside down (Figure 6): the higher-pressure air beside the car flows around the edge toward the low pressure beneath it (Section 3). The strength of the resulting vortex depends on how large this pressure difference is and over what length of edge it acts. A floor-edge vortex is therefore strongest where the suction is strongest, around the diffuser throat, and weak at low speed, where all pressure differences are small (Section 2).

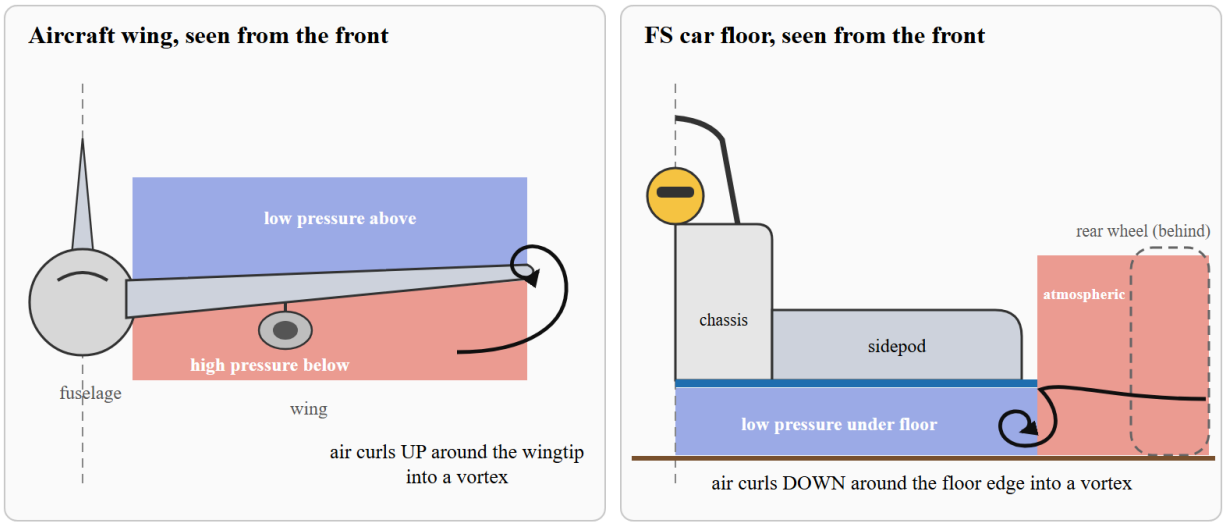


Figure 6. A wingtip and a floor edge form vortices by the same mechanism, upside down. Schematic only.

The shape of the edge matters a great deal (Figure 7). When air flows around a sharp edge, it cannot follow the sudden change of direction: it separates exactly at the edge, leaving as a thin sheet of rotating air that rolls up into a vortex. Because the separation point is fixed by the geometry, the vortex forms in the same place, with a predictable strength, over a wide range of conditions. Around a rounded edge, the air can follow the curve for some distance before separating, so the separation point is no longer fixed. Wind-tunnel tests on aircraft wings confirm this: rounding the edge delays the formation of the vortex and shifts its position, and under some conditions no vortex forms at all[13]. In short, a sharp edge forces a vortex to form, whereas a smooth, rounded edge merely allows one to form under favorable conditions.

What matters is the size of the rounding compared with the flow around it: a small radius still behaves almost like a sharp edge, whereas a curve comparable in size to the gap beneath the floor lets the air follow it. This is the main reason why the smooth channel on the UFV floor is unlikely to create a strong vortex on its own. Its outer wall does end in a thin lip at floor level, where the flow must separate and a smaller vortex may form; however, the curved wall has already turned the air gradually before it reaches the lip, so this vortex is likely to be weaker than one shed from a plain sharp edge, and its actual strength can only be established by testing. The FS rules do not prevent a sharp edge in this sense: the minimum radii they require, 3 mm on forward-facing edges and 1 mm elsewhere (T 2.4.1)[1], are small compared with the 25 mm

channel.

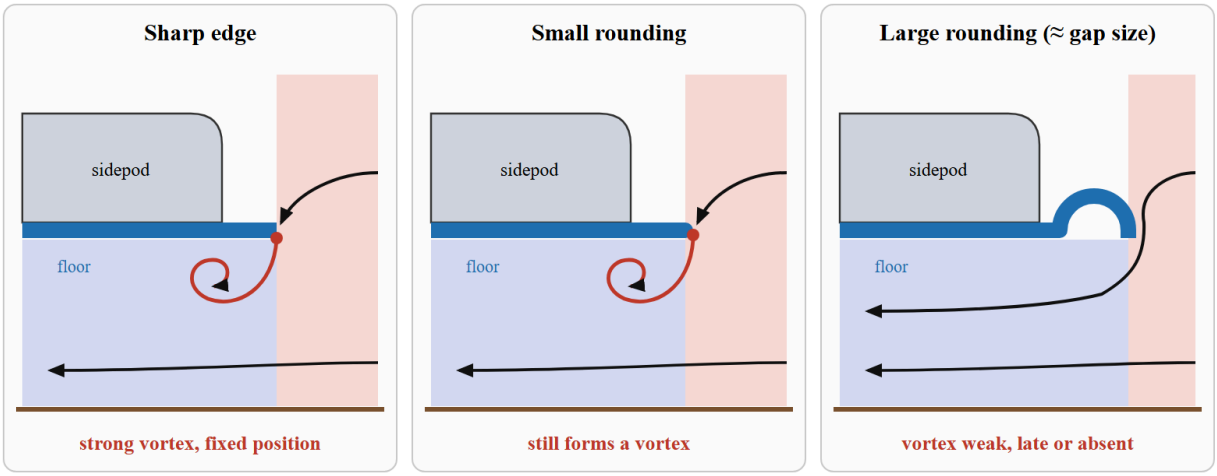


Figure 7. The same floor with different edges: a sharp edge and a small rounding both form a vortex, whereas a large rounding may not. The current UFV halfpipe (bottom) is rounded on the outside but ends in a thin lip, where a smaller vortex of uncertain strength may form. Schematic only.

Once formed, a vortex does not last indefinitely. Friction within the air gradually spreads out and weakens the rotation as the vortex travels downstream. More abruptly, a vortex can break down: when it travels into a region where the pressure rises, its core slows, swells and collapses into a large, disorganized region of turbulence (Figure 8). This is precisely the condition inside a diffuser, where the pressure rises toward the exit (Section 2), and vortex breakdown is one of the mechanisms identified by Mahon *et al.* and by Ruhrmann and Zhang for the sudden loss of downforce at very low ride heights[6],[7]. A vortex can also be weakened or destroyed by external disturbances, such as turbulent air from the wake of a front tire mixing into it, or by striking a solid object such as a rear tire. Creating a vortex strong enough to seal the edge, and keeping it intact along the full length of the floor, is therefore the central design problem addressed in Section 6.

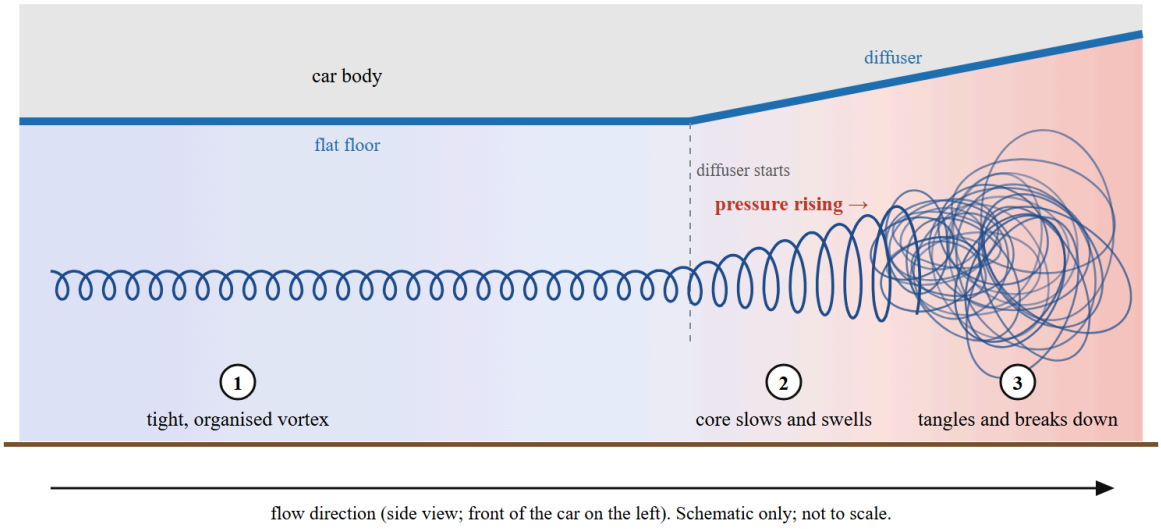
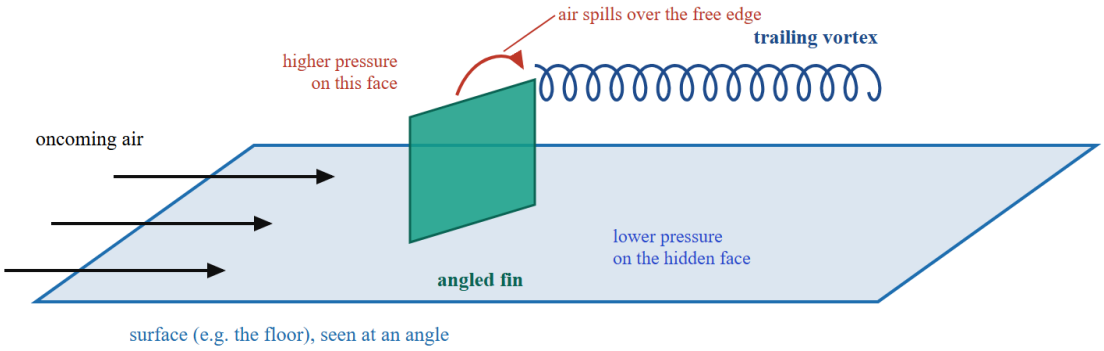


Figure 8. A vortex swelling and breaking down as it travels into the rising pressure of a diffuser. Schematic only.

6. Vortex Generators and the Virtual Skirt

For a vortex to act as a virtual skirt, four conditions must be met. It must be created deliberately, ideally from a sharp edge that fixes where it forms. It must be strong enough to resist the inflow, which means the device that creates it must be sized accordingly. It must be positioned along the part of the edge where leakage is greatest. Finally, it must survive along the floor: it must be protected from the wake of the front tire, kept from breaking down in the rising pressure of the diffuser, and steered clear of the rear tire. This section describes the devices that can create such a vortex and assesses how well each meets these conditions.

The simplest of these devices is the vortex generator: a small, thin fin set at an angle to the oncoming air (Figure 9). Because it is angled, the fin behaves like a tiny wing. The air on one face is slightly slowed and the air on the other face slightly accelerated, so a pressure difference forms across the fin. At its free edge, air flows around from the higher-pressure face to the lower-pressure face and, exactly as at a wingtip or a floor edge (Section 5), rolls up into a vortex that trails downstream. The vortex becomes stronger as the fin's angle, its size and the speed of the air increase, up to the point where the angle is so large that the flow separates from the fin's surface, an effect called stall, after which the vortex becomes weaker and less organized. Vortex generators of this kind are widely used on aircraft and are also used on race cars to increase downforce[2].



The angled fin acts like a small wing: air flows around its free edge from the higher- to the lower-pressure face and rolls up. Schematic only.

Figure 9. How a vortex generator works: an angled fin acts as a small wing, and air spilling around its free edge rolls up into a trailing vortex. Schematic only.

On aircraft, vortex generators are sized relative to the boundary layer, the thin layer of slow air next to the surface (Section 2). Conventional devices are roughly as tall as the boundary layer, while so-called low-profile devices, with heights of only 10–50% of the boundary-layer thickness, have proved effective at preventing flow separation[14]. Those devices, however, are designed to mix fast air into the slow layer, not to seal a gap. Sealing a floor edge requires a vortex large enough to act across the gap beneath the floor, which points toward devices at the larger end of this range. The FS rules also limit their size: any device fitted beneath the floor counts toward the 30 mm minimum ground clearance (T 2.2.1)[1], so a fin that protrudes below the floor forces the floor itself to sit correspondingly higher.

The direction in which the vortex spins also matters. A vortex running along the floor edge drives the air around it in a circle. If it spins so that, on its side closest to the ground, the air is driven outwards, it opposes the air trying to flow in under the floor and may even carry air out from beneath it. If it spins the other way, it assists the inflow instead. The direction of rotation is set by the way the device is angled, so the orientation of every device must be chosen with this in mind.

Vane-type vortex generators are small, flat fins mounted near the front of the floor edge or along it, each angled to the flow (Figure 10). Several can be placed in a row so that their vortices combine into one stronger vortex, as in the floor-edge vanes of modern Formula One cars, where the first vanes create the vortices and the following ones strengthen and guide them[11]. Vanes are light, cheap and simple to make from flat composite or aluminum sheet. Their angle sets both the strength of the vortex and the direction in which it travels, and, mounted ahead of the existing channel, they could feed a vortex into it, which the channel's concave shape

may then help to hold in place. Their weaknesses are that, being small, they sit in the turbulent wake of the front tire, and that each must pass the 50 N load test that applies to every aerodynamic device in FS (T 8.3.2)[1].

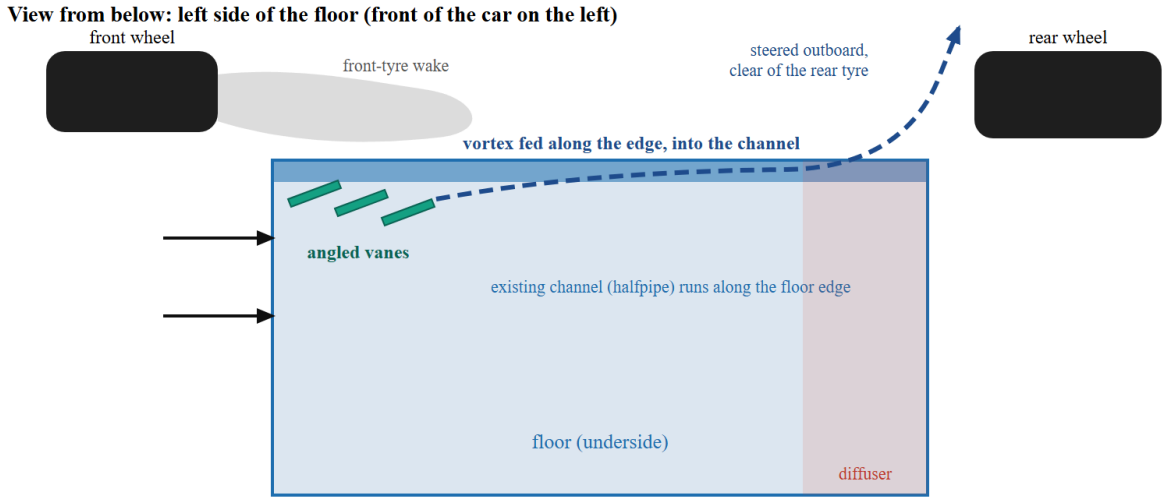


Figure 10. Vane-type vortex generators near the front of the floor edge, feeding a vortex along the edge. Schematic only.

Strakes are longer, thin vertical fins running lengthwise beneath the floor, for example near the start of the diffuser (Figure 11). Their sharp lower edges shed vortices along their length and help to hold those vortices in a fixed position. Because they are mounted inboard of the floor edge, however, their vortices act on the flow beneath the floor rather than at the edge itself, and, since they protrude downwards, their depth is limited by the 30 mm ground clearance.

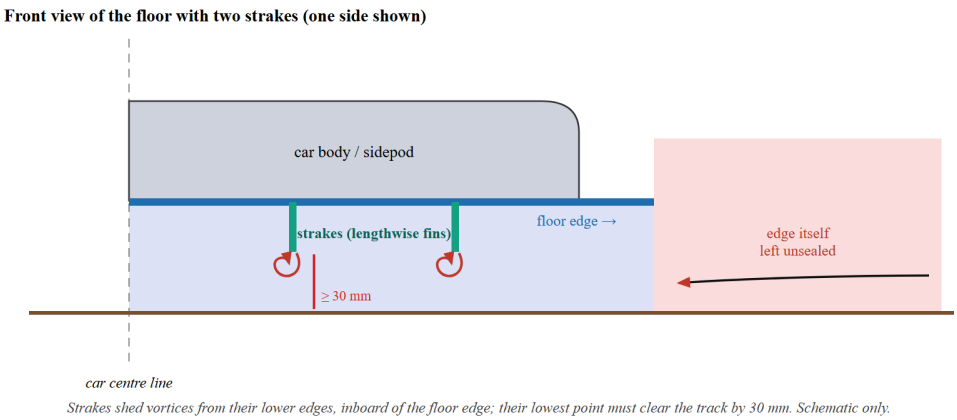


Figure 11. Strakes: lengthwise fins beneath the floor that shed vortices from their lower edges. Schematic only.

A floor edge wing is a small, lengthwise airfoil running just outboard of the floor edge (Figure 12). The 2022 Formula One regulations provide for one explicitly, positioned between 5 and 20 mm from the floor; its purpose is to direct air upwards and outwards at the floor edge, keeping it from re-entering beneath the car, an effect described as pseudo-sealing[15]. An edge wing offers the greatest control over where the air goes, but it is the most complex of these devices to design and build, it must pass both the 200 N and 50 N load tests (T 8.3.1, T 8.3.2)[1], and it must stay within the car's width limit (T 8.2.2)[1].

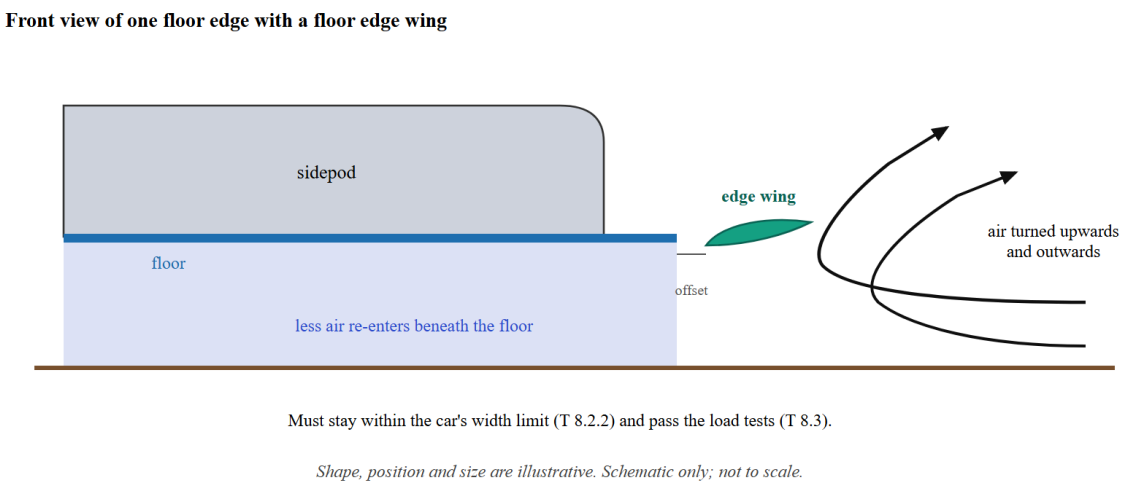


Figure 12. A floor edge wing: a small lengthwise airfoil just outboard of the floor edge, turning the air upwards and outwards. Schematic only.

Finally, a fence or downturned lip is a thin vertical plate along the floor edge, extending downwards toward the track (Figure 13). Its sharp lower edge fixes the point where the air separates, so it produces a well-defined edge vortex, and it also narrows the opening through which air can leak. Like any other part, it must still clear the track by 30 mm (T 2.2.1)[1], so it acts as a lowered sharp edge rather than a skirt, and the rest of the floor must sit correspondingly higher. Because it forms an angled section with the floor, it also stiffens the floor edge, which makes it a candidate replacement for the channel in the alternative design examined in Section 9.

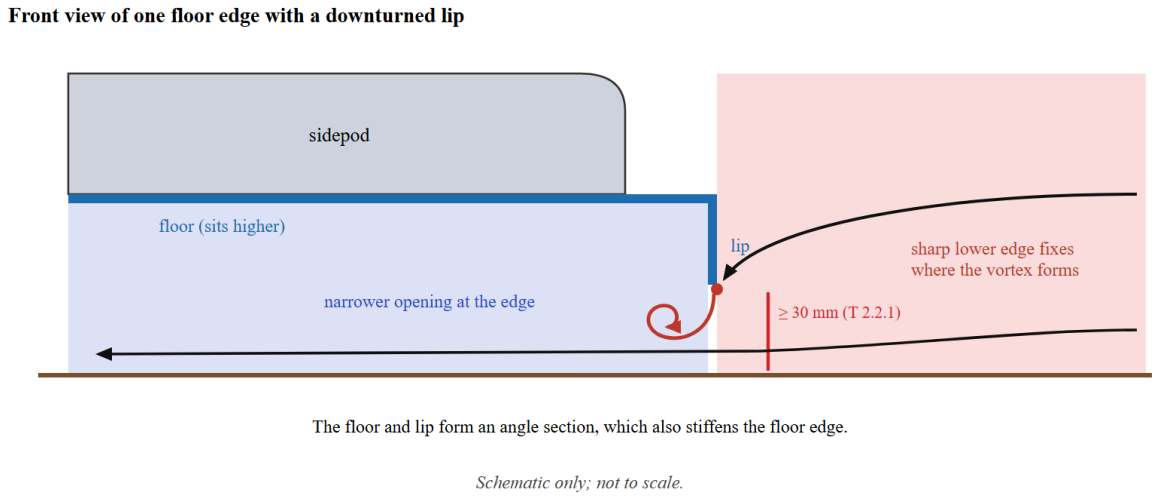


Figure 13. A downturned lip at the floor edge: its sharp lower edge fixes where the vortex forms, and it stiffens the edge. Schematic only.

Table 1 compares the four devices against the criteria that matter most for the UFV car. The assessment is qualitative: it is a judgment based on the principles and sources discussed above, not on measured data, and it is intended to identify which devices merit testing rather than to rank them precisely.

Device	Works with the halfpipe	Robust to front-tire wake	Steers vortex clear of rear tire	Low drag	Fits FS rules easily	Easy to build
Vane-type vortex generators	✓	~	✓	✓	~	✓
Strakes	~	~	~	✓	~	✓
Floor edge wing	~	~	✓	~	~	✗
Downturned lip	✗ (replaces it)	~	✗	✓	~	✓

Table 1. Qualitative comparison of floor-edge vortex devices for the UFV car: ✓ good, ~ partial, ✗ poor.

Two candidates stand out. Vane-type vortex generators score well on almost every criterion, are the simplest to build and can work together with the existing channel, so they form the basis of the main proposal in Section 8. A downturned lip is the natural choice for the alternative design without the channel, because it creates a vortex and stiffens the edge at the same time, and it is examined in Section 9. Strakes and edge wings are not ruled out, but they

either act away from the edge or demand considerably more design and manufacturing effort. Before either proposal can be detailed, however, the constraints specific to Formula Student must be considered, which is the subject of Section 7.

7. Constraints Specific to Formula Student

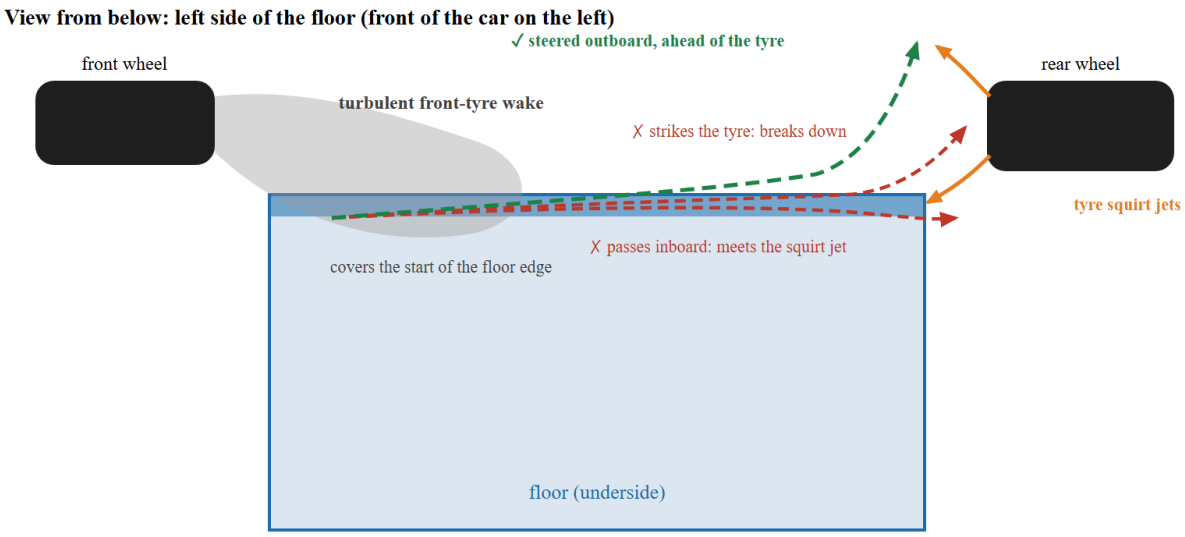
The devices described in Section 6 were developed largely for aircraft and for Formula One cars, which operate at far higher speeds, with larger budgets and under different rules. Before any of them can be proposed for the UFV car, the conditions under which an FS floor actually works must be considered: its low speed, the tires that surround it, the way the car moves in corners, the competition rules and the resources of a student team.

The first difference is speed. At 48 km/h, every pressure difference around the car is small (Section 2). To a first approximation this does not, by itself, prevent a virtual skirt from working, because both the strength of an edge vortex and the strength of the inflow it must resist grow with the dynamic pressure in the same way. What low speed does change is the behavior of the air close to the surfaces. At the low Reynolds number of an FS car, the slow layers of air next to the floor are relatively thick and separate more readily (Section 2), so a small device can sit partly within this slow air and produce a weaker vortex than its size suggests. The absolute gains are also modest, because every force on the car scales with the same small dynamic pressure; on a car whose total downforce is itself small, however, even a modest gain is a meaningful fraction.

The second constraint comes from the front tires. A rotating wheel produces a large, turbulent wake that is carried rearwards along the side of the car[4], passing directly over the front of the floor edge, which is exactly where a vortex generator would naturally be placed. A vortex that forms inside this disturbed air starts weak, and a wake that sweeps across an existing vortex can disrupt it (Section 5). Devices must therefore either be placed where the flow is cleaner or be large enough for their vortex to dominate the surrounding turbulence. The position of the wake also changes with steering, so a device that sits in clean air when the car travels straight may sit in the wake in a corner.

At the other end of the floor, the rear tires pose the opposite problem. Where each rear tire meets the track, it squeezes air out sideways in the jets known as tire squirt (Section 4), and the inboard jet flows toward the edge of the diffuser[10],[11]. An edge vortex that travels the length of the floor arrives exactly in this region. If it strikes the tire, it breaks down; if it passes inboard

of the tire, it meets the squirt jet head on. Formula One teams address this by shaping the flow so that it joins the tire squirt outside the rear tire[11]. For the UFV car, controlling where the vortex goes at the rear wheel is therefore as important as creating it at the front (Figure 14).



Possible paths of the edge vortex at the rear wheel. Schematic only; not to scale.

Figure 14. The front-tire wake disturbs the start of the floor edge, and the rear tire and its squirt jets lie at its end: the vortex must be created in clean air and steered clear of the rear tire. Schematic only.

Corners add a further complication. An FS car spends most of each lap turning, and in a corner it travels at a small angle to the oncoming air, known as yaw, so the two floor edges are no longer symmetrical: one faces the oncoming flow more directly, while the other is partly sheltered (Section 3). At the same time, the body rolls toward the outside of the corner, reducing the gap beneath the outer floor edge and increasing it beneath the inner one (Figure 15). The leakage, the strength of the edge vortex and its path will therefore differ between the two sides of the car, and a design that works well in a straight line must also be checked at the yaw and roll angles typical of a corner.

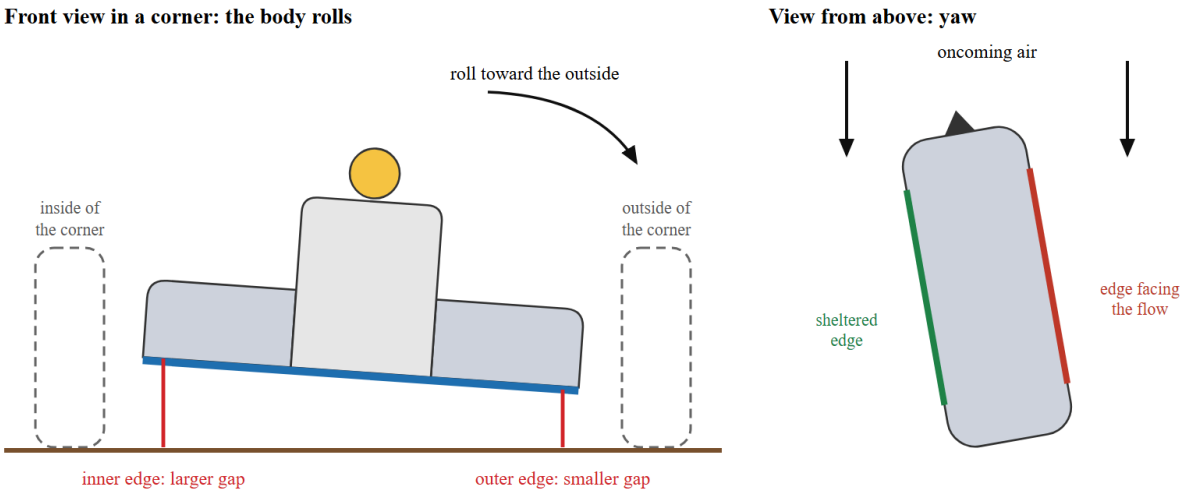


Figure 15. In a corner the car rolls and yaws: the gap shrinks under the outer floor edge and grows under the inner one, so the two edges seal differently. Schematic only.

The competition rules then set firm limits on what can be built. No part of the car other than the tires may come closer than 30 mm to the track (T 2.2.1), and no aerodynamic device may touch it (T 2.2.2), so every device must stay above the gap it is trying to seal. Any part designed to guide the airflow counts as an aerodynamic device (T 8.1.1) and must therefore stay within the car's width limit (T 8.2.2) and withstand the 200 N and 50 N load tests (T 8.3.1, T 8.3.2), while its edges must respect the minimum radii of 3 mm and 1 mm (T 2.4.1)[1]. None of these rules prevents the devices of Section 6, but together they limit their size and position.

As noted in Section 4, exhaust blowing remains legal for a combustion-engined car such as the UFV car[1]. Several factors nevertheless make it unattractive. First, the jet exists only while the engine is under load: when the driver lifts off the throttle, the seal weakens and the downforce changes mid-corner, making the car less predictable. Formula One teams overcame this with engine maps that kept the exhaust flowing off-throttle[10], but doing so burns fuel, which FS penalizes through its efficiency scoring. Second, the available flow is small: the 20 mm intake restrictor (CV 1.7.2)[1] limits the air an FS engine can consume, especially at the part throttle typical of slow corners. Third, the exhaust gas is several hundred degrees hot, so the adjacent composite floor and diffuser would need heat shielding, and any rerouted exhaust must still satisfy the 110 dB(C) noise limit (CV 3.2.1)[1]. Finally, an outlet shaped specifically to direct airflow could be judged an aerodynamic device under T 8.1.1[1]. For these reasons, this paper treats exhaust blowing as a possible future development rather than a primary solution.

Finally, an FS team works with limited time, money and manufacturing capability, and its floor is regularly damaged by curbs, cones and bottoming. Any device should therefore be simple to make, light, cheap to replace and easy to remove, and, because its best size and angle cannot be known in advance, it should ideally be adjustable, for example a bolt-on vane with several mounting positions. These requirements favor the simple vanes and lip identified in Section 6, which are developed into concrete proposals in Sections 8 and 9.

8. Proposal A: Vortex Generators on the Floor Edge

This section applies the principles of Sections 5 and 6 to the floor of the UFV car. Each side of the floor is a flat panel 936.75 mm long and approximately 274 mm wide. The sidepod sits on its inboard part and covers less than half of its width, while the 25 mm channel runs along its outer edge, with its open front end immediately behind the front tire. The ride height is set at the 30 mm minimum allowed by the rules (T 2.2.1)[1], and neither the ride height nor the floor itself can be changed to accommodate a new device. Every device considered here must therefore add nothing below the bottom surface of the floor.

This requirement rules out the most direct arrangement, in which vanes hang beneath the floor and act on the leaking air itself. A vane mounted there would become the lowest point of the car, and the 30 mm clearance would then be measured from its tip rather than from the floor, so the whole floor would have to be raised by the height of the vane, sacrificing part of the suction that the device is meant to protect (Section 2). Two arrangements remain that leave the underside untouched: vanes on the upper surface of the floor, which form the main proposal, and vanes inside the channel, which are presented as an alternative later in this section.

The upper surface offers a clear strip of floor between the sidepod and the channel (Figure 16). The vanes are proposed for the outer part of this strip, immediately inboard of the channel, for three reasons. First, they must not sit directly in front of the sidepod, where they would disturb the air entering its inlet and could reduce cooling. Second, they should be placed behind the densest part of the front-tire wake, which passes over the front corner of the floor (Section 7), so that their vortex forms in cleaner air. Third, they must still be far enough forward for the vortex to form fully and travel along most of the floor, where it can act on the leakage. These requirements point to a starting position close to the front of the sidepod, as shown in Figure 16; the exact position is one of the variables to be studied.

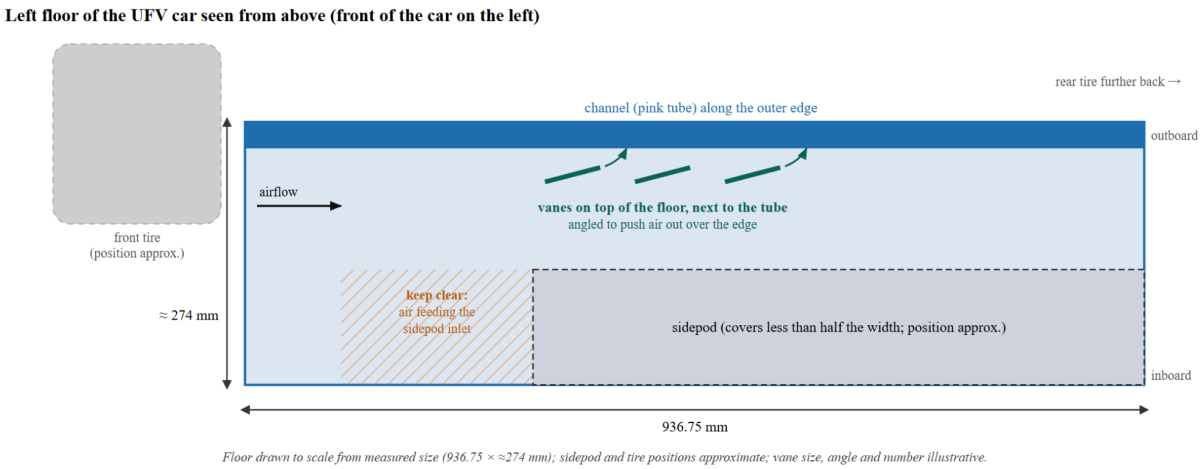


Figure 16. Proposed position of the vanes on the left floor of the UFV car, seen from above: on top of the floor next to the channel, behind the front-tire wake and clear of the sidepod inlet. Floor to scale; sidepod and tire positions approximate.

Each vane is a small, flat, upright plate set at a slight angle to the flow, so that it turns the air outwards, toward the edge. Like the tip of a wing (Section 5), the vane then sheds a vortex from its upper edge, as air spills over it from its outboard face, where the pressure is higher, to its inboard face, where it is lower. With the vane angled this way, the vortex spins so that its lower side, closest to the floor, moves outwards (Figure 17). This is the direction of spin identified in Section 6 as the one that opposes the inflow near the ground.

The direction of spin also determines where the vortex goes. A vortex traveling close to a flat surface does not stay in place: because the air cannot flow through the surface, the vortex drifts sideways along it, in the same direction as the air moving on its side closest to the surface, as if pushed by a mirror image of itself on the other side of the wall[12]. For the vortex shed by these vanes, whose lower side moves outwards, this drift carries it toward the edge of the floor rather than inboard toward the sidepod. The vane angle therefore does two jobs at once: it gives the vortex the correct spin, and it moves the vortex toward the place where it is needed. Whether the vortex then passes over the channel and settles beside the edge, at the height of the gap, or remains above the floor cannot be predicted from this reasoning alone, and it is the first question that the study described at the end of this section must answer.

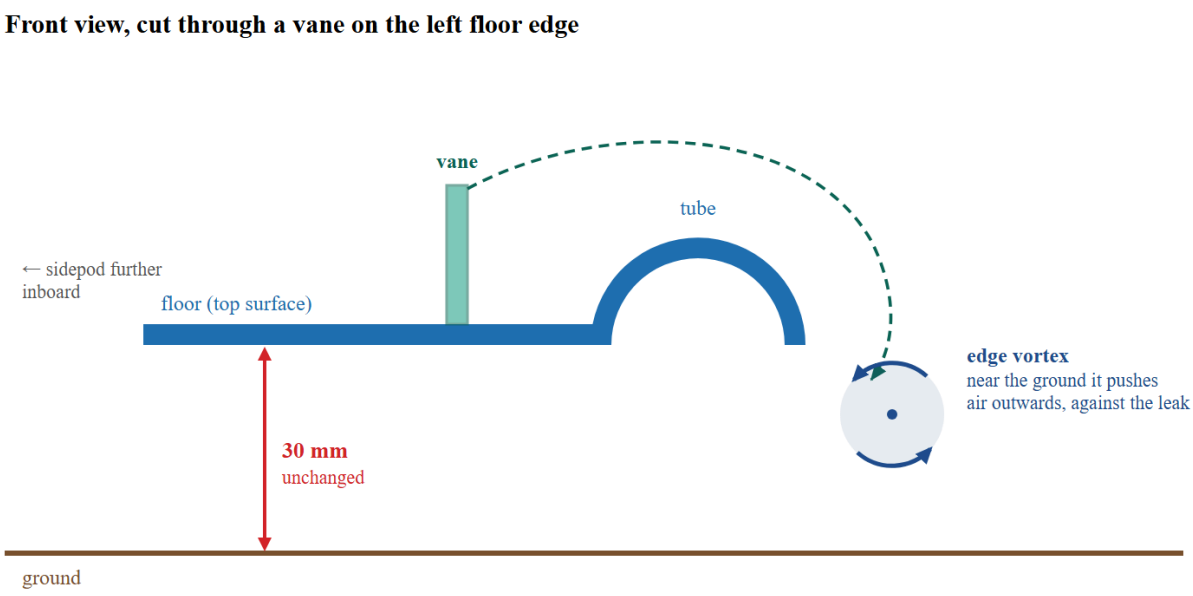


Figure 17. Front view through a vane: the vortex shed from its top spins so that its lower side moves outwards, which also makes it drift toward the floor edge, where it opposes the inflow near the ground. Floor, channel and gap to scale; vortex position illustrative.

The number of vanes involves a trade-off. Each vane adds swirl, and since all the vanes are angled the same way, their vortices spin in the same direction. A numerical study of pairs of vanes found that vortices spinning in the same direction tend to wind around each other and merge with little loss of strength, whereas vortices of opposite spin weakened each other by up to 30%[16]; all the vanes in the row must therefore be angled the same way. Several vanes can thus build a stronger vortex than one, and the Aston Martin Formula One floor, for example, uses groups of vanes along its edge: three at the front to set up vortices, followed by further groups that turn the flow on the floor outboard into them, all combining to guide the flow toward the tire squirt[11]. More vanes are not, however, automatically better. Spacing also matters: in experiments on rows of vortex generators, placing them closer together produced stronger vortices at first but made them decay faster downstream[17]. Each vane also adds drag, which at 48 km/h may cancel much of the small gain in downforce, and each is one more part exposed to damage from cones and curbs (Section 7). Finally, the space is limited: the vanes must lie behind the front-tire wake but far enough forward to leave the vortex enough floor to act on, and on a floor 936.75 mm long this window allows only a few vanes at sensible spacing.

For these reasons, this paper does not fix the number of vanes. Instead, it proposes

mounting provisions for up to five vanes, for example a row of threaded inserts in the floor, five being a practical limit for the space available. The number should then be built up one vane at a time: starting with a single vane, a second is added and the result compared, and so on, stopping when an additional vane no longer improves the downforce enough to justify its drag. The angle and height of the vanes should be treated in the same way, through several fixing positions, starting from the sizing guidance of Section 6[14].

At the rear of the floor, the vortex must not be carried into the diffuser, where the rising pressure would cause it to break down (Section 5), nor into the rear tire and its squirt jets (Section 7, Figure 14). Because the vortex drifts outwards as it travels, it may already pass outboard of the rear tire without further help; if the study shows that it does not, a small turning vane at the rear end of the floor edge can be added to direct it outboard, ahead of the tire.

An alternative that also leaves the underside untouched is to place the vanes inside the channel itself (Figure 18). The channel is open toward the ground and rises about 12.5 mm above the bottom of the floor, so small vanes fixed to its roof can hang into it without protruding below the floor. Because its open front end faces the oncoming air, the channel carries air along the edge, and vanes inside it would twist this air into a vortex exactly at the edge, without relying on the drift described above. The option has three drawbacks: the vanes must be small to fit, which limits the strength of their vortex; they partly block the channel; and it is not known how much air actually flows through the channel, so the vanes may have little air to work with. The real internal dimensions of the channel, and any fixings inside it, would also have to be checked before a vane size could be chosen. This option is therefore kept as a secondary candidate, to be evaluated alongside the upper-surface vanes.

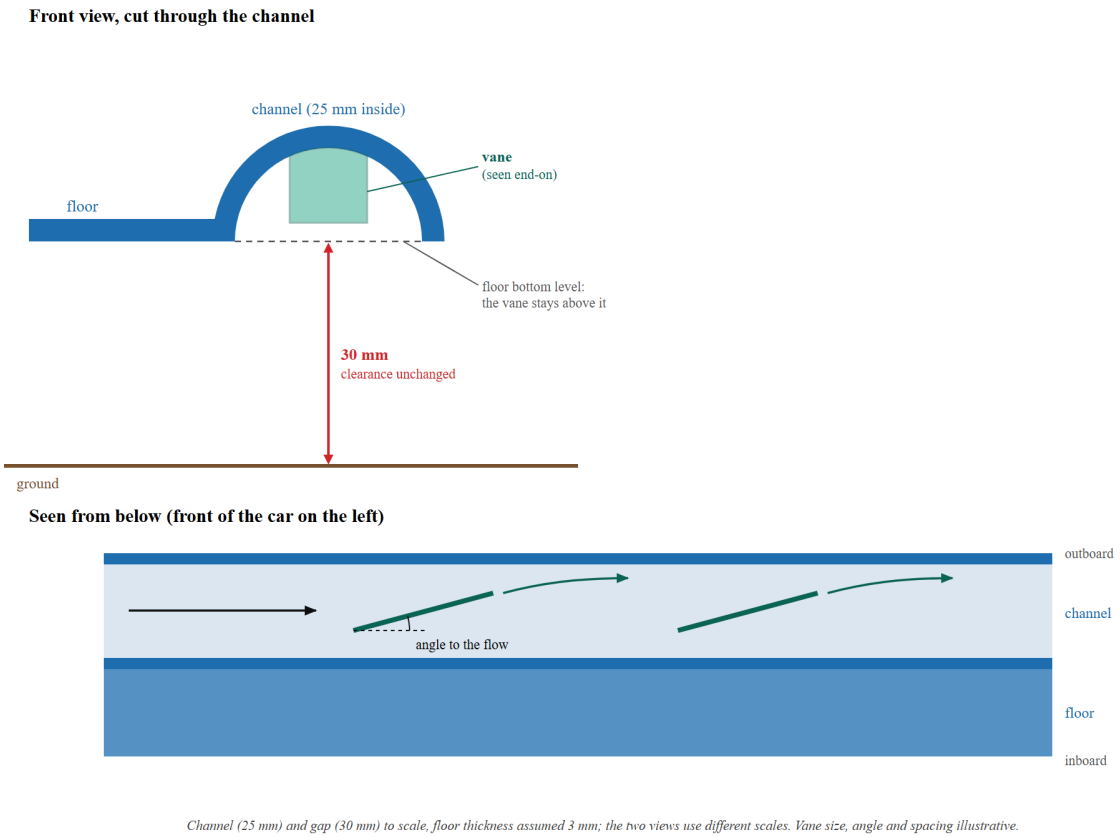


Figure 18. The alternative: vanes fixed to the roof of the channel, shown in a front cut (top) and from below (bottom). They stay above the bottom of the floor, so the 30 mm clearance is unchanged. Channel and gap to scale; vane size and angle illustrative.

A track test requires the car, a test day and most of the team, so it is not a practical way of exploring the many combinations of vane position, angle, height and number. This exploration should therefore be carried out in CFD, using the team's existing simulation model of the car, and each configuration compared with the current floor. The study should first establish whether the vortex forms and where it travels, using cross-sections of the flow at several stations along the floor edge, and then compare the floor's downforce and the car's drag. The best configurations should also be checked at the yaw and roll angles of a typical corner (Section 7), and the flow into the sidepod inlet checked to ensure that cooling is not affected. Only the most promising configuration would then be fitted to the car and confirmed during a test session that is already planned, using simple methods such as flow-visualization paint or wool tufts, which show the direction of the flow on the floor without requiring a dedicated test.

If the study confirms the reasoning of this section, Proposal A offers a way to reduce leakage along the floor edges with small, inexpensive, bolt-on parts, without changing the ride

height, the floor or the sidepods. Its main risks are that the vortex remains above the floor instead of reaching the gap, that the front-tire wake weakens it, or that it is steered into the rear tire; each of these would be visible in the CFD results, and an unsuccessful configuration can simply be removed. Section 9 examines the more radical alternative of replacing the channel altogether.

9. Proposal B: Replacing the Channel with a Sharp Lip

The second proposal takes a different approach. Instead of adding devices to the existing floor, it changes the floor edge itself, replacing the rounded channel with a shape designed to create an edge vortex on its own. As in Section 8, the ride height and the main floor surface stay where they are; only the outer strip of the floor changes.

The two proposals differ above all in what they ask of the team. Proposal A works with the floor built for last year's car: the vanes are bolted onto it, can be adjusted or removed, and leave the floor itself untouched. Proposal B cannot be applied in this way. The channel must be removed, either by cutting it from the existing floor and rebuilding the edge or by making a new floor, a change that cannot be undone and that costs considerably more time and work. In exchange, Proposal B addresses the cause identified in Section 5, the rounded shape of the edge, rather than adding parts to compensate for it.

Section 6 described the downturned lip as a thin plate along the floor edge, extending toward the track. On the UFV car, however, the floor already sits at the 30 mm minimum (T 2.2.1)[1], so a lip extending below it is not allowed, and lowering the lip would mean raising the whole floor. Proposal B avoids this by raising only the outer strip of the floor. Approaching the edge, the floor curves gradually upwards and then drops sharply into a vertical lip whose tip returns to the level of the floor's bottom surface (Figure 19). The lowest point of the car, and therefore the 30 mm clearance, is unchanged, and the new edge occupies roughly the same strip of floor as the current channel.

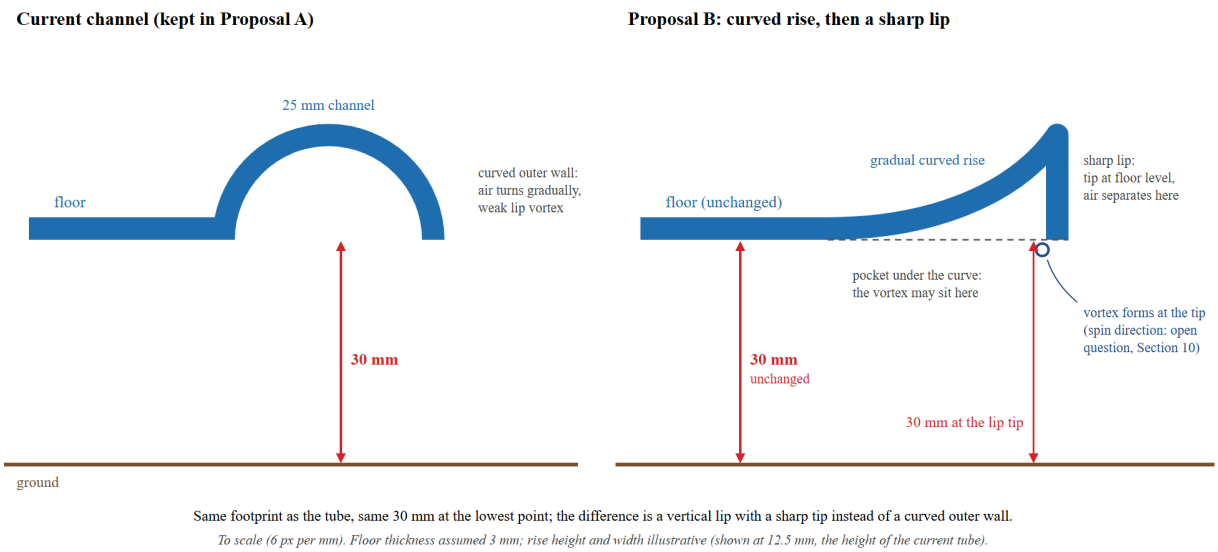


Figure 19. The current channel (left) and Proposal B (right), drawn to scale: the same footprint and the same 30 mm clearance, but a gradual curved rise ending in a vertical lip with a sharp tip instead of a curved outer wall. Rise height illustrative.

Seen this way, Proposal B is not a radical departure from the current design. Both are a raised strip along the edge, open toward the ground and ending in a lip at floor level. The difference lies in the shape of the outer wall. On the channel, the curved wall turns the air gradually before it reaches the lip, so any vortex formed there is weak (Section 5). On Proposal B, the outer face is a flat vertical plate, so the air flowing down it must separate exactly at the tip, which fixes the position of the vortex and makes its strength more predictable. The minimum edge radii required by the rules (T 2.4.1)[1] are small compared with the 30 mm gap, so the tip still behaves as a sharp edge. Only the tip needs to be sharp: the upper corner, where the curve meets the lip, can be rounded to simplify manufacture.

A curved rise is preferred to a square step for three reasons. First, it raises less of the floor: the gap beneath the raised strip is larger than elsewhere, and the suction there correspondingly lower (Section 2), so the strip should be kept no wider than the current channel. Second, it avoids sharp internal corners on the underside, in which slow-moving air could collect. Third, it leaves a smooth pocket directly behind the lip, in which the edge vortex may settle and be held in place, the role originally expected of the channel; whether it does so is one of the questions that the CFD study described in Section 8 must answer.

The new edge must also take over the structural role of the channel, which helps to stiffen the floor edge. Because the curved rise and the lip together form a folded section, in the same way that folding a sheet of card along a line makes it much stiffer, the new edge should provide

comparable stiffness; its exact laminate and thickness are a structural question outside the scope of this paper.

One question remains open. With vanes, the angle of each vane sets the direction in which the vortex spins (Section 8). A plain lip offers no such control: the only air flowing around its tip is the air leaking in under the floor, so the vortex it forms is created by the inflow itself and spins in the same direction as the inflow, with its side closest to the ground moving inwards. Such a vortex cannot push air back out, as described in Section 6. It can still help, however: its low-pressure core sits in the gap and partly blocks it, in the manner of the virtual skirts described in Section 4[5]. How much sealing this blocking alone can provide is discussed in Section 10.

The raised edge also affects the air on top of the floor. Rising above the floor's upper surface, it acts as a low wall that prevents air flowing along the top of the floor from spilling outwards over the edge, and guides it rearwards, toward the rear tire. This is not a new effect, since the current channel rises by a similar amount, but it should not be made worse. The edge vortex also runs rearwards and must be released outboard of the rear tire, for the reasons given in Section 7 (Figure 14). Both issues can be addressed at the rear end of the floor, either by tapering the rise down to the flat floor so that the air on top can spill outwards, or by turning the lip outboard so that it releases the vortex ahead of the tire.

Proposal B is therefore a larger commitment than Proposal A, but a more fundamental one: it gives the floor an edge designed to create a vortex while keeping the clearance, the footprint and the stiffness of the current design. The two proposals are also compatible, since the vanes of Proposal A could be fitted to the upper surface of a Proposal B floor. Section 10 compares the two and recommends how the team should proceed.

10. Discussion and Recommendation

Sections 8 and 9 presented two ways of creating a virtual skirt on the UFV floor without changing its ride height. Table 2 compares them on the points that matter most to the team. As with Table 1, the comparison is qualitative: it summarizes the reasoning of the previous sections rather than measured results.

Criterion	Proposal A: vanes on top	Proposal B: sharp lip
Change to the current floor	None; vanes bolted on	Channel removed; edge rebuilt or new floor
Reversible	Yes	No
Ride height	Unchanged	Unchanged
Control of vortex spin	Set by the vane angle	Set by the inflow
Where the vortex forms	Above the floor; must drift to the gap	At the gap, fixed by the sharp tip
Adjustability	Position, angle and number	Fixed once built
Cost and time	Low	Moderate to high

Table 2. Qualitative comparison of Proposal A (vanes on the upper surface of the existing floor) and Proposal B (curved rise and sharp lip replacing the channel).

The table shows that the two proposals are complementary rather than competing: the main weakness of each is the main strength of the other. Proposal A controls the direction in which the vortex spins, because the angle of each vane sets it, but its vortex forms above the floor and must drift over the channel to reach the gap, which it may not do (Section 8). Proposal B forms its vortex exactly at the gap, at a sharp edge that fixes its position, but it has no control over the spin, which is set by the inflow itself (Section 9). Formula One floors use both approaches: shaped cut-outs in the floor edge to initiate the vortices that form virtual skirts[5], and vanes that turn the surrounding air into those vortices and strengthen them[11]. A floor that combines a sharp lip with vanes on its upper surface may therefore perform better than either proposal alone.

The cost of the two proposals is very different, however, and this determines the order in which they should be pursued. Evaluating either proposal in CFD costs only time, since it requires nothing more than a change to the model; building them does not. Proposal A can be fitted to last year's floor at little cost and removed if it fails, whereas Proposal B requires the floor edge to be rebuilt or a new floor to be made. The recommendation of this paper is therefore a staged approach. First, both proposals, and their combination, should be modeled in CFD alongside the current floor, following the study described in Section 8. Second, if Proposal A improves the floor in CFD, it should be fitted to the existing car and confirmed during a planned test session. Third, Proposal B should be considered for the next floor the team designs, where it

can be built in from the start rather than added to an existing part, but only if the CFD study shows that the sharp lip, alone or with vanes, seals the floor better than the current channel.

The most important open question is how much a vortex formed by the inflow itself can seal the floor. A vortex created by vanes can be made to spin so that it pushes air back out from under the floor, whereas one created by a plain lip spins with the inflow and can only partly block the gap (Section 9). The Formula One evidence suggests that such blocking is worthwhile[5], but it comes from cars at far higher speeds and is reported in technical journalism rather than in measured data. Comparing Proposal B with and without vanes in the CFD study would answer this question directly for the UFV car.

This study also has several limitations. No simulation or measurement of the UFV car has been presented: the proposals rest on established principles and on results obtained mainly for aircraft wings and Formula One cars, at Reynolds numbers far higher than those of an FS car[13], where the slower, thicker layers of air near the surfaces may weaken small devices (Section 7). Much of the Formula One evidence comes from technical journalism[5],[11], which describes what teams have built but not how much it gains. The floor dimensions used in this paper were measured on the car, but the positions of the sidepod and tires in the figures are approximate, and the internal dimensions of the channel were not measured. The behavior of the vortex as the ride height changes over bumps, under braking and in corners was considered only qualitatively, the stiffness of the new edge in Proposal B was not analyzed, and no estimate of the gain in downforce has been made. Each of these points can be addressed by the CFD study and by measurements of the car.

Within these limits, the reasoning of this paper points to a clear course of action: evaluate both proposals in CFD, fit the inexpensive vanes of Proposal A to the current floor if they prove effective, and keep the sharp lip of Proposal B as the basis for the next floor if the simulations support it. Section 11 summarizes the conclusions of the paper.

11. Conclusion

This paper has examined how the floor of the UFV Racing Team's Formula Student car could be sealed at its edges by vortices rather than by solid skirts. At the typical speed of 48 km/h and the minimum ride height of 30 mm, the floor loses part of its suction to air drawn in from the sides, and the rules prevent this gap from being closed physically. The rounded channel

along the current floor edge is unlikely to create a strong edge vortex on its own, because its curved outer wall lets the air turn gradually instead of forcing it to separate at a sharp edge. Sealing the floor with a virtual skirt therefore requires a vortex that is created deliberately, positioned along the edge and steered clear of the front-tire wake and the rear tire.

Two proposals were developed that achieve this without changing the ride height or the main floor. Proposal A adds small vanes to the upper surface of the existing floor, next to the channel. Their angle sets the direction in which the vortex spins and also makes it drift toward the floor edge, and their number is built up one at a time rather than fixed in advance. Proposal B replaces the channel with a gradual curved rise ending in a sharp lip at floor level, which fixes the position of the vortex at the gap and keeps the edge stiff, but leaves the direction of spin to the inflow. The two are complementary: Proposal A controls the spin, Proposal B controls the position, and a floor combining both may perform better than either alone.

Because Proposal A can be fitted to last year's floor at little cost while Proposal B requires the edge to be rebuilt, the paper recommends a staged approach: evaluate both proposals and their combination in CFD, fit the vanes of Proposal A to the current car if they prove effective, and build the sharp lip of Proposal B into the next floor if the simulations support it. The main uncertainties are whether the vortex of Proposal A reaches the gap, how well a vortex formed by the inflow seals it, and how much either gains at the low speeds of Formula Student; all three can be answered by the CFD study described in Section 8.

More generally, this paper shows that the edge vortex of an FS floor need not be left to chance. By applying established principles to the measured geometry and the rules of a specific car, a student team can design the vortex it wants, decide where it forms and where it goes, and test the result at low cost before committing to a new part.

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